DDA3020 Machine Learning

Tutorial: Support Vector Machine

April 2026

Instructions

This tutorial covers the main ideas of Support Vector Machines (SVM), including large margin classification, hinge loss, Lagrange duality and KKT conditions, SVM with slack variables, and SVM with kernels. Show your working clearly for all calculation questions.

Useful Formulas

For a binary classification dataset $D = \{(\mathbf{x}_i, y_i)\}_{i=1}^m$ with $\mathbf{x}_i \in \mathbb{R}^n$, $y_i \in \{-1, +1\}$:

- Decision function: $f_{\mathbf{w},b}(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$
- Distance from $\mathbf{x}$ to hyperplane: $\frac{|f_{\mathbf{w},b}(\mathbf{x})|}{\|\mathbf{w}\|}$
- Hard-margin SVM primal: $\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \forall i$
- Hinge loss: $\max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b))$
- Dual problem: $\max_{\boldsymbol{\alpha}} \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \quad \text{s.t.} \quad \sum_i \alpha_i y_i = 0,\ \alpha_i \geq 0$
- Primal solution from dual: $\mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$
- Soft-margin SVM: $\min_{\mathbf{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i,\ \xi_i \geq 0$
- Kernel: $k(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^\top \phi(\mathbf{x}_j)$
- RBF kernel: $k(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$

Exercises

Q1. Which of the following statements about SVM is correct?

(A) SVM minimizes the training error without any regularization

(B) The decision boundary of SVM is determined by all training data points

(C) SVM finds the hyperplane that maximizes the margin between classes

(D) SVM requires the data to be linearly separable in all cases

(Ref: Stanford CS229, Andrew Ng's SVM notes; Lecture slides pp. 4–11)

Q2. Given the hyperplane $\mathbf{w}^\top \mathbf{x} + b = 0$ with $\mathbf{w} = [3, 4]^\top$ and $b = -10$, compute the distance from the point $\mathbf{x}_0 = [1, 2]^\top$ to this hyperplane.

(Ref: Lecture slides pp. 15–17; Bishop PRML §7.1)

Q3. Which of the following is NOT a property of support vectors in a hard-margin SVM?

(A) They lie on the margin boundaries $y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$

(B) They have $\alpha_i > 0$ in the dual solution

(C) Removing any non-support-vector training point does not change the decision boundary

(D) They are the points farthest from the decision boundary

(Ref: Lecture slides p. 37; CMU 10-701 SVM exercises)

Q4. Consider a 2D dataset with three points: $\mathbf{x}_1 = [0, 0]^\top$ with $y_1 = -1$, $\mathbf{x}_2 = [2, 2]^\top$ with $y_2 = +1$, $\mathbf{x}_3 = [2, 0]^\top$ with $y_3 = +1$.

(a) Find the optimal separating hyperplane $\mathbf{w}^\top \mathbf{x} + b = 0$ by solving the hard-margin SVM problem.

(b) Identify the support vectors.

(c) Compute the margin.

(Ref: Stanford CS229 Problem Sets; Bishop PRML Exercise 7.1)

Q5. Consider the hard-margin SVM primal problem:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1,\ \forall i$$

Derive the Lagrangian and show that the stationarity conditions yield:

$$\mathbf{w} = \sum_{i=1}^{m} \alpha_i y_i \mathbf{x}_i \quad \text{and} \quad \sum_{i=1}^{m} \alpha_i y_i = 0$$

(Ref: Lecture slides pp. 33–34; Boyd & Vandenberghe, Convex Optimization, Ch. 5)

Q6. Starting from the Lagrangian of the hard-margin SVM and the stationarity conditions in Q5, derive the dual problem:

$$\max_{\boldsymbol{\alpha}} \sum_{i=1}^{m} \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \quad \text{s.t.} \quad \sum_i \alpha_i y_i = 0,\ \alpha_i \geq 0, \forall i$$

(Ref: Lecture slides pp. 35–36; Andrew Ng CS229 notes)

Q7. Explain the role of the complementary slackness condition $\alpha_i(1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)) = 0$ in the SVM solution. What does it tell us about the relationship between α_i and the position of $\mathbf{x}_i$ relative to the margin?

(Ref: Lecture slides p. 37; KKT conditions review, Boyd & Vandenberghe Ch. 5)

Q8. In the soft-margin SVM with slack variables, the primal problem is:

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{m} \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i,\ \xi_i \geq 0,\ \forall i$$

(a) Write down the Lagrangian function with dual variables $\alpha_i \geq 0$ and $\mu_i \geq 0$.

(b) Derive the stationarity condition with respect to ξ_i and show that $\alpha_i = C - \mu_i$.

(c) Show that in the dual problem, the only change compared to the hard-margin dual is the constraint $0 \leq \alpha_i \leq C$.

(Ref: Lecture slides pp. 43–46; Bishop PRML §7.1.1)

Q9. In the soft-margin SVM solution, explain the geometric meaning of the three cases: $\alpha_i = 0$, $0 < \alpha_i < C$, and $\alpha_i = C$. For the case $\alpha_i = C$, what is the significance of $\xi_i \leq 1$ versus $\xi_i > 1$?

(Ref: Lecture slides p. 47; CMU 10-701 SVM tutorial)

Q10. What happens to the SVM decision boundary as $C \to \infty$ in the soft-margin formulation? What happens as $C \to 0$? Explain intuitively.

(Ref: Stanford CS229; Lecture slides pp. 25–26, 43)

Q11. Consider a 1D dataset: $x_1 = -1$ ($y_1 = -1$), $x_2 = 0$ ($y_2 = -1$), $x_3 = 1$ ($y_3 = +1$), $x_4 = 2$ ($y_4 = +1$). This data is linearly separable.

(a) Find the optimal hard-margin SVM solution (w, b).

(b) Which points are support vectors?

(c) What is the margin?

(Ref: MIT 6.034 SVM exercises; Lecture slides pp. 18–22)

Q12. Consider the two points $\mathbf{x}_1 = [1, 0]^\top$ and $\mathbf{x}_2 = [0, 1]^\top$ in $\mathbb{R}^2$. Define the feature mapping $\phi(\mathbf{x}) = [x_1^2, \sqrt{2}\, x_1 x_2, x_2^2]^\top$.

(a) Compute $\phi(\mathbf{x}_1)^\top \phi(\mathbf{x}_2)$.

(b) Verify that $k(\mathbf{x}_1, \mathbf{x}_2) = (\mathbf{x}_1^\top \mathbf{x}_2)^2$ gives the same result.

(c) What is the advantage of using the kernel k instead of explicitly computing ϕ?

(Ref: Lecture slides pp. 51–53; Andrew Ng CS229 kernel notes)

Q13. For the RBF (Gaussian) kernel $k(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$:

(a) What is $k(\mathbf{x}, \mathbf{x})$ for any $\mathbf{x}$?

(b) As $\sigma \to 0$, describe the behavior of the kernel and the resulting decision boundary.

(c) As $\sigma \to \infty$, what does the kernel approach, and what does this imply for the SVM classifier?

(Ref: Lecture slides p. 54; Bishop PRML §6.2)

Q14. Compare the hinge loss used in SVM with the log loss used in logistic regression. Both are plotted as functions of $y_i(\mathbf{w}^\top \mathbf{x}_i + b)$. Explain why the hinge loss leads to a sparse solution (in terms of support vectors) while the log loss does not.

(Ref: Lecture slides p. 60; Stanford CS229)

Q15. A student has a dataset with $n = 10{,}000$ features and $m = 100$ training samples. Another student has $n = 5$ features and $m = 500$ samples. For each scenario, recommend whether to use (i) logistic regression / linear SVM, or (ii) SVM with RBF kernel. Justify your answer.

(Ref: Lecture slides p. 61; Andrew Ng's practical advice on SVM)

Solutions

Q1. Answer: C.

Explanation: SVM finds the hyperplane that maximizes the margin between the two classes. Option A is wrong because SVM includes a regularization term $\frac{1}{2}\|\mathbf{w}\|^2$. Option B is wrong because the decision boundary depends only on support vectors (points with $\alpha_i > 0$), not all training points. Option D is wrong because soft-margin SVM with slack variables and kernel SVM can handle non-separable data.

Q2. Answer: $\frac{1}{5}$

Explanation: The distance is $\frac{|\mathbf{w}^\top \mathbf{x}_0 + b|}{\|\mathbf{w}\|} = \frac{|3 \cdot 1 + 4 \cdot 2 - 10|}{\sqrt{3^2+4^2}} = \frac{|3+8-10|}{\sqrt{25}} = \frac{1}{5}$.

Q3. Answer: D.

Explanation: Support vectors are the points *closest* to the decision boundary (lying exactly on the margin), not the farthest. They satisfy $y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$, which corresponds to distance $\frac{1}{\|\mathbf{w}\|}$ from the hyperplane. Options A, B, and C are all correct properties.

Q4. Answer:

(a) By symmetry and inspection, the optimal hyperplane passes between the negative point and the two positive points. We need $\mathbf{w}$ and b such that: $y_1(\mathbf{w}^\top \mathbf{x}_1 + b) = 1 \Rightarrow -(b) = 1 \Rightarrow b = -1$. The closest positive point determines the other margin constraint. Testing: the support vectors are $\mathbf{x}_1 = [0,0]^\top$ and $\mathbf{x}_3 = [2,0]^\top$ (both on the line connecting them along the x_1-axis). By the margin constraints: $-(w_1 \cdot 0 + w_2 \cdot 0 + b) = 1$ and $w_1 \cdot 2 + w_2 \cdot 0 + b = 1$, giving $b = -1$ and $2w_1 - 1 = 1$ so $w_1 = 1$. We also need $w_1 \cdot 2 + w_2 \cdot 2 + b = 2w_2 + 1 \geq 1$, so $w_2 \geq 0$. To minimize $\frac{1}{2}\|\mathbf{w}\|^2 = \frac{1}{2}(w_1^2 + w_2^2)$, set $w_2 = 0$. Thus $\mathbf{w} = [1,0]^\top$, $b = -1$, and the hyperplane is $x_1 = 1$.

(b) Support vectors: $\mathbf{x}_1 = [0,0]^\top$ and $\mathbf{x}_3 = [2,0]^\top$ (both satisfy the margin constraint with equality). $\mathbf{x}_2 = [2,2]^\top$ gives $y_2(2-1) = 1$, so it is also on the margin and is a support vector.

(c) Margin $= \frac{2}{\|\mathbf{w}\|} = \frac{2}{1} = 2$.

Q5. Answer: The Lagrangian is:

$$\mathcal{L}(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\|\mathbf{w}\|^2 + \sum_{i=1}^{m} \alpha_i \big(1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)\big)$$

Setting $\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \mathbf{0}$:

$$\mathbf{w} - \sum_{i=1}^{m} \alpha_i y_i \mathbf{x}_i = \mathbf{0} \quad \Rightarrow \quad \mathbf{w} = \sum_{i=1}^{m} \alpha_i y_i \mathbf{x}_i$$

Setting $\frac{\partial \mathcal{L}}{\partial b} = 0$:

$$-\sum_{i=1}^{m} \alpha_i y_i = 0 \quad \Rightarrow \quad \sum_{i=1}^{m} \alpha_i y_i = 0$$

Q6. Answer: Substituting $\mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$ and $\sum_i \alpha_i y_i = 0$ into $\mathcal{L}$:

$$\begin{aligned}
\mathcal{L} &= \frac{1}{2}\left\|\sum_i \alpha_i y_i \mathbf{x}_i\right\|^2 + \sum_i \alpha_i - \sum_i \alpha_i y_i \left(\sum_j \alpha_j y_j \mathbf{x}_j\right)^\top \mathbf{x}_i - b\underbrace{\sum_i \alpha_i y_i}_{=0} \\
&= \frac{1}{2}\sum_{i,j} \alpha_i\alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j + \sum_i \alpha_i - \sum_{i,j} \alpha_i\alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \\
&= \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i\alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j
\end{aligned}$$

Maximizing over $\boldsymbol{\alpha}$ with constraints $\alpha_i \geq 0$ and $\sum_i \alpha_i y_i = 0$ gives the dual.

Q7. Answer: The complementary slackness condition states that for each i, either $\alpha_i = 0$ or $y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$ (or both). If a point lies strictly outside the margin, i.e., $y_i(\mathbf{w}^\top \mathbf{x}_i + b) > 1$, then α_i must be zero, meaning it does not contribute to $\mathbf{w}$. Only points exactly on the margin ($y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$) can have $\alpha_i > 0$; these are the support vectors that determine the classifier.

Q8. Answer:

(a) $\mathcal{L} = \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \xi_i + \sum_i \alpha_i(1 - \xi_i - y_i(\mathbf{w}^\top \mathbf{x}_i + b)) + \sum_i \mu_i(-\xi_i)$

(b) $\frac{\partial \mathcal{L}}{\partial \xi_i} = C - \alpha_i - \mu_i = 0 \;\Rightarrow\; \alpha_i = C - \mu_i$.

(c) Since $\alpha_i \geq 0$ and $\mu_i \geq 0$, we have $0 \leq \alpha_i \leq C$. After substitution, the ξ_i terms vanish from the dual objective (since $C - \alpha_i - \mu_i = 0$), giving the same dual form as before but with $0 \leq \alpha_i \leq C$.

Q9. Answer: $\alpha_i = 0$: the point is correctly classified and lies outside the margin; it does not influence the classifier. $0 < \alpha_i < C$: then $\mu_i > 0$, so $\xi_i = 0$ by complementary slackness ($\mu_i \xi_i = 0$). The point lies exactly on the margin and is correctly classified. $\alpha_i = C$: then $\mu_i = 0$, so $\xi_i \geq 0$. The point is inside the margin or misclassified. If $\xi_i \leq 1$, the point is on the correct side of the decision boundary but inside the margin. If $\xi_i > 1$, the point crosses the decision boundary and is misclassified.

Q10. Answer: As $C \to \infty$, the penalty for violations becomes infinite, so the SVM enforces all constraints strictly and behaves like a hard-margin SVM (if the data is separable). As $C \to 0$, violations are barely penalized, so the margin becomes very wide and many points may be misclassified; the classifier becomes underfitting with a very large margin.

Q11. Answer:

(a) By symmetry, the optimal boundary is at $x = 0.5$. The support vectors are $x_2 = 0$ ($y = -1$) and $x_3 = 1$ ($y = +1$). The constraints give $-(w \cdot 0 + b) = 1$ and $w \cdot 1 + b = 1$, so $b = -1$ and $w = 2$. The hyperplane is $2x - 1 = 0$, i.e., $x = 0.5$.

(b) Support vectors: $x_2 = 0$ and $x_3 = 1$. (Check: $x_1 = -1$ gives $y_1(2(-1) - 1) = (-1)(-3) = 3 > 1$ and $x_4 = 2$ gives $y_4(2(2) - 1) = 3 > 1$, so they are not support vectors.)

(c) Margin $= \frac{2}{|w|} = \frac{2}{2} = 1$.

Q12. Answer:

(a) $\phi(\mathbf{x}_1) = [1, 0, 0]^\top$, $\phi(\mathbf{x}_2) = [0, 0, 1]^\top$. Thus $\phi(\mathbf{x}_1)^\top \phi(\mathbf{x}_2) = 0$.

(b) $k(\mathbf{x}_1, \mathbf{x}_2) = (\mathbf{x}_1^\top \mathbf{x}_2)^2 = (1 \cdot 0 + 0 \cdot 1)^2 = 0$. Same result.

(c) The kernel computes the inner product in the high-dimensional feature space without explicitly constructing $\phi(\mathbf{x})$. This is crucial when ϕ maps to a very high (or even infinite) dimensional space, avoiding the computational cost of working directly in that space. This is known as the "kernel trick."

Q13. Answer:

(a) $k(\mathbf{x}, \mathbf{x}) = \exp(0) = 1$ for any $\mathbf{x}$.

(b) As $\sigma \to 0$, $k(\mathbf{x}_i, \mathbf{x}_j) \to 0$ for $\mathbf{x}_i \neq \mathbf{x}_j$ and $k(\mathbf{x}_i, \mathbf{x}_i) = 1$. Each training point only influences predictions in its immediate vicinity. The decision boundary becomes very irregular and overfits the training data.

(c) As $\sigma \to \infty$, $k(\mathbf{x}_i, \mathbf{x}_j) \to 1$ for all pairs. All points look equally similar, so the kernel matrix approaches the all-ones matrix. The SVM cannot distinguish between points and produces a nearly linear (or trivial) classifier, leading to underfitting.

Q14. Answer: Both hinge loss $\max(0, 1 - y_i f(\mathbf{x}_i))$ and log loss $\log(1 + e^{-y_i f(\mathbf{x}_i)})$ are convex upper bounds on the 0–1 loss. However, the hinge loss is exactly zero when $y_i f(\mathbf{x}_i) \geq 1$, meaning correctly classified points sufficiently far from the boundary contribute zero loss. By complementary slackness, these points have $\alpha_i = 0$ and do not affect the solution at all, yielding sparsity. The log loss, by contrast, is always strictly positive, so every training point exerts some influence on the parameters, and no sparsity arises.

Q15. Answer:

- **Scenario 1** (n = 10,000, $m = 100$): With far more features than samples, the data is likely linearly separable in the original space. Use logistic regression or linear SVM (no kernel). A kernel SVM could easily overfit with so few samples.
- **Scenario 2** ($n = 5$, $m = 500$): With few features and moderate data, the data may not be linearly separable. SVM with RBF kernel is appropriate, as it can capture non-linear boundaries. The number of samples is manageable for kernel methods (the kernel matrix is 500×500).